

EXPERIMENT – 38

Implementation of Known Plaintext Attack on Hill Cipher

CODE:

```
#include <stdio.h>

int modInverse(int a)
{
    for (int i = 1; i < 26; i++)
        if ((a * i) % 26 == 1)
            return i;
    return -1;
}

int main()
{
    int P[2][2], C[2][2], K[2][2];
    int det, inv, i, j;

    printf("Enter 2 plaintext-ciphertext pairs (4 numbers each):\n");

    printf("Plaintext matrix P:\n");
    for (i = 0; i < 2; i++)
        for (j = 0; j < 2; j++)
            scanf("%d", &P[i][j]);

    printf("Ciphertext matrix C:\n");
    for (i = 0; i < 2; i++)
```

```

    for (j = 0; j < 2; j++)
        scanf("%d", &C[i][j]);

/* P inverse */
det = (P[0][0] * P[1][1] - P[0][1] * P[1][0]) % 26;
if (det < 0) det += 26;

inv = modInverse(det);

if (inv == -1)
{
    printf("Attack failed: P has no inverse modulo 26.\n");
    return 0;
}

/* K = C * P^-1 */
K[0][0] = inv * (P[1][1] * C[0][0] - P[0][1] * C[1][0]);
K[0][1] = inv * (P[1][1] * C[0][1] - P[0][1] * C[1][1]);
K[1][0] = inv * (-P[1][0] * C[0][0] + P[0][0] * C[1][0]);
K[1][1] = inv * (-P[1][0] * C[0][1] + P[0][0] * C[1][1]);

printf("\nRecovered Hill Cipher Key:\n");

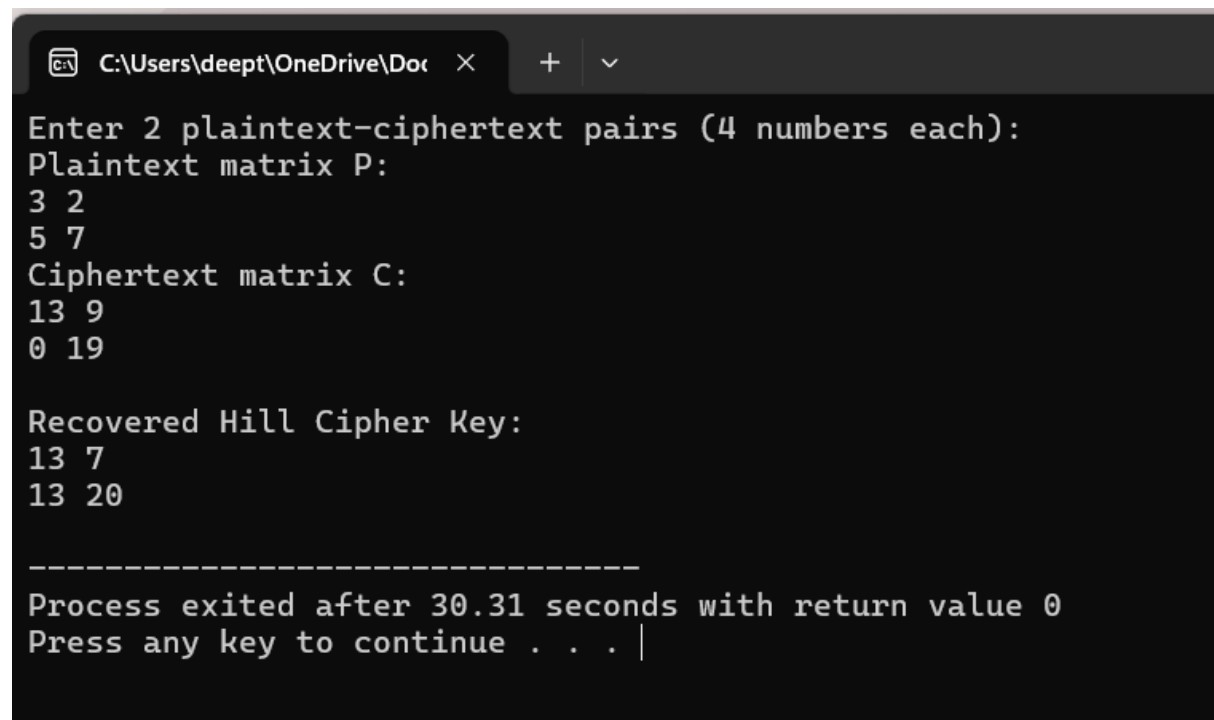
for (i = 0; i < 2; i++)
{
    for (j = 0; j < 2; j++)
        printf("%d ", ((K[i][j] % 26) + 26) % 26);

    printf("\n");
}

```

```
    return 0;  
}
```

OUTPUT:



```
C:\Users\deept\OneDrive\Doc × + v  
Enter 2 plaintext-ciphertext pairs (4 numbers each):  
Plaintext matrix P:  
3 2  
5 7  
Ciphertext matrix C:  
13 9  
0 19  
  
Recovered Hill Cipher Key:  
13 7  
13 20  
  
-----  
Process exited after 30.31 seconds with return value 0  
Press any key to continue . . . |
```